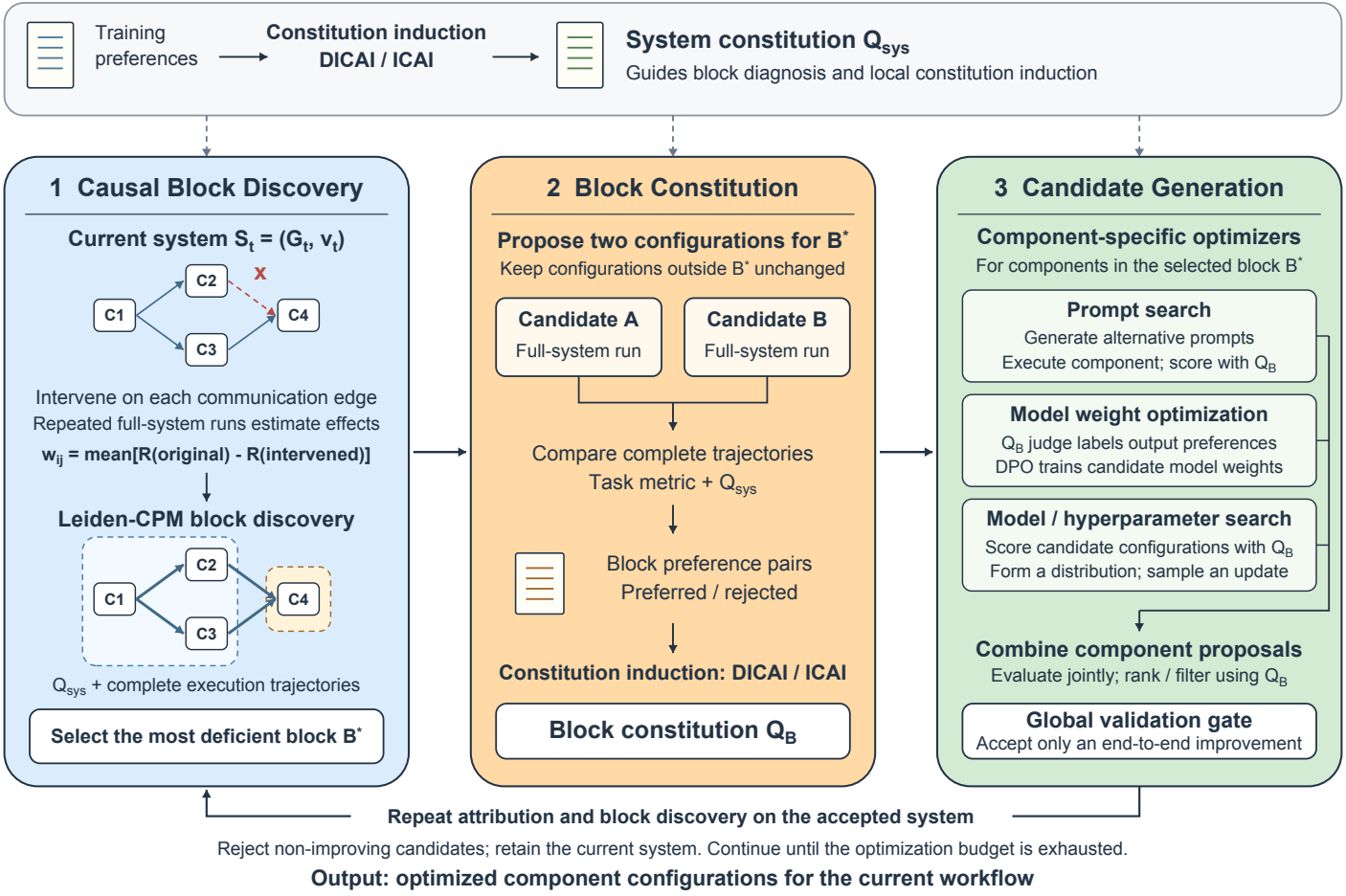

Training preferences
Constitution induction
DICAI / ICAI
System constitution Q_{sys}
Guides block diagnosis and local constitution induction
1 Causal Block Discovery
Current system $S_t = (G_t, v_t)$
C1
C2
C3
C4
X
Intervene on each communication edge
Repeated full-system runs estimate effects
w_{ij} = mean[R(original) - R(intervened)]
Leiden-CPM block discovery
Q_{sys} + complete execution trajectories
Select the most deficient block B*
2 Block Constitution
Propose two configurations for B*
Keep configurations outside B* unchanged
Candidate A
Full-system run
Candidate B
Full-system run
Compare complete trajectories
Task metric + Q_{sys}
Block preference pairs
Preferred / rejected
Constitution induction: DICAI / ICAI
Block constitution Q_B
3 Candidate Generation
Component-specific optimizers
For components in the selected block B*
Prompt search
Generate alternative prompts
Execute component; score with Q_B
Model weight optimization
Q_B judge labels output preferences
DPO trains candidate model weights
Model / hyperparameter search
Score candidate configurations with Q_B
Form a distribution; sample an update
Combine component proposals
Evaluate jointly; rank / filter using Q_B
Global validation gate
Accept only an end-to-end improvement
Repeat attribution and block discovery on the accepted system
Reject non-improving candidates; retain the current system. Continue until the optimization budget is exhausted.
Output: optimized component configurations for the current workflow